Questions and Answers with Distinguished Guest Dr. Lisa Su, President and CEO of Advanced Micro Devices, Inc. (AMD)

Dr. Lisa T. Su is AMD President and Chief Executive Officer, a position she has held since October 2014, and serves on the AMD Board of Directors.

Prior to joining AMD, Dr. Su served as senior vice president and general manager, Networking and Multimedia, at Freescale Semiconductor, Inc. (a semiconductor manufacturing company), and was responsible for global strategy, marketing, and engineering for the company's embedded communications and applications processor business.

From 1994 to 2007, Dr. Su spent 13 years at IBM in various engineering and business leadership positions, including vice president of the Semiconductor Research and Development Center responsible for the strategic direction of IBM's silicon technologies, joint development alliances and semiconductor RD operations.

Dr. Su earned bachelor's, master's and doctorate degrees in electrical engineering from the Massachusetts Institute of Technology (MIT). She has published more than 40 technical articles and was named a Fellow of the Institute of Electronics and Electrical Engineers in 2009. In 2018, Dr. Su was elected to the National Academy of Engineering and received the Global Semiconductor Association's Dr. Morris Chang Exemplary Leadership Award. In 2019, Dr. Su was named one of the World's 50 Best CEOs by Barron's, one of the "Most Powerful Women in Business" by Fortune Magazine, and she was included in "The Bloomberg 50," a list of people who defined the year. In 2020, Fortune named Dr. Su #2 on its "Business Person of the Year" list, she was *elected to the American Academy of Arts Science, received the Grace Hopper Technical Leadership Abie Award, and was honored by the Semiconductor Industry Association with its highest honor, the Robert N. Noyce Award. Dr. Su has been a member of the board of directors of Cisco Systems, Inc., since January 2020. She also serves on the board of directors for the Semiconductor Industry Association. (Advanced Micro Devices)*

Dr. Su's answers are taken from an interview conducted by Josephine Lee and Mr. Kevin Anderson from AMD and have been edited for clarity.

1) Who were your great mentors when you were a "kid" and how did they help you?

My mom's been a large inspiration in my life – my parents' sacrifice and commitment to excellence were key drivers for me. The pushed me to realize my full potential. And they never expected anything from me that they didn't expect from themselves.

2) What building-block experiences should teenagers who are interested in STEM look for?

I like to believe that my training as an engineer is about how to solve problems. How can you analytically solve a problem and make that fun? And that is one of the key parts of making STEM interesting, especially to women and underrepresented minorities.

3) How did growing up in NYC influence your love for science and technology and your outlook on life?

As a child, I spent time taking apart my brother's broken remote-control cars and repairing them. That curiosity and determination to fix an issue helped pave my way to be interested in engineering and technology.

In addition, I was very fortunate to have attended the Bronx High School of Science. Bronx Science students take a college preparatory curriculum that includes four years of lab science, math, English, and social studies. We also had the opportunity to do independent research.

When I was in college, the hottest majors were electrical engineering and computer science. Electrical engineering dealt with hardware, and I wanted to actually build things. Most people also said it was the "harder" major and I thought I would definitely learn a lot. I learned when I chose something very difficult and did well, it would give me great confidence for the next challenge.

4) What advice would you give your 16-year-old self?

One, a good, strong education is very important. Nobody can take your education away from you; it is your calling card. Two, be ambitious in what you do. And, three, take feedback to heart and learn from your failures. I really believe that's what makes you better.

I was given a piece of advice when I was a young engineer at IBM that has really stuck with me. Someone told me, "You should run toward problems." If you're going to work on something, work on something that's really important.

5) Why do you think women are underrepresented in STEM-related fields? How have you tried to address this gap?

Exposure to STEM subjects is extremely important to young people. There are incredible opportunities for women and minorities. It is also important that you can see the fruits of your work and their impact on society.

At AMD, we're especially passionate about enabling the imagination and creativity of the next generation. Technology in their hands encourages exploration, imagination and learning – opening doors to new careers and possibilities. That's why we're partnering with local organizations in four cities to establish AMD-powered learning labs to inspire students to pursue STEM education. AMD is providing hardware and other support in each new lab, as well as ongoing engagement from our employee volunteers. These four AMD Learning Labs are expected to impact more than 1,000 students.

Education provides incredible opportunity for students to expand their horizons and achieve their fullest potential. And with access to powerful technology in the classroom, they may be the future innovators, researchers, leaders or entrepreneurs to dream up new ways to solve society's challenges.

6) As an executive who has to make long-term decisions today, how should students learn the skill of making long-term decisions?

When I took over as CEO at AMD, I felt that what we needed to do was focus on our core competency, which for AMD is high-performance computing. This meant setting a 5-year vision and roadmap for how we would build the best products in the industry. We laid out a clear strategy based on focus and disciplined execution and asked the whole organization to align around them. This also meant deciding not to do some things, like not pursuing some markets that were good markets, but just not in our core competency. Deciding what not to do is as important as deciding what to do.

7) Many of today's engineers choose software over hardware. What are the advantages of choosing a career in hardware?

A software and hardware career can both be very rewarding for the right person. In hardware, you get to "take things apart," examine, fix, and put it back together again.

It is one of the fields where you can make a difference by the choices you make on the technology front.

8) How do you see the semiconductor industry impacting our lives five years from now?

This is a very unique time in the semiconductor market in which I think people are just realizing how important chips are to every part of our lives. That's why I am in this business. It's really humbling to know that the chips we are building are so important no matter who you are and what you are trying to do.

In technology, particularly in semiconductors, we have to make bets far in advance. Decisions made today will impact products that come to market three or four years from now. We do spend a lot of time with our customers learning what problems they are facing so that we can develop new solutions. We also spend time studying overall technology trends, identifying the big bottlenecks, and key innovations that are on the horizon.